

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458392.824 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.004 +/- 0.051
Transit depth (Rp/Rs)^2	0.0012 +/- 0.036 [%]
Orbital Inclination (inc)	85.11 +/- 1.71
Airmass coefficient 1 (a1)	2.08 +/- 1.36
Airmass coefficient 2 (a2)	-0.49 +/- 0.45
Scatter in the residuals of the lightcurve fit is	63.56 %
Best Comparison Star	#3 - [418, 142]
Optimal Aperture	1.11
Optimal Annulus	5.5
Transit Duration (day)	0.032 +/- 0.035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.004 ± 0.051 vs 0.131 (deviation 2.49σ)
Duration fit vs published	0.032 d vs 0.1075 d (deviation 2.16σ)
Mid-time O–C	-4.4 min
Depth SNR	0.1
Residual scatter	63.56 %

Light curve

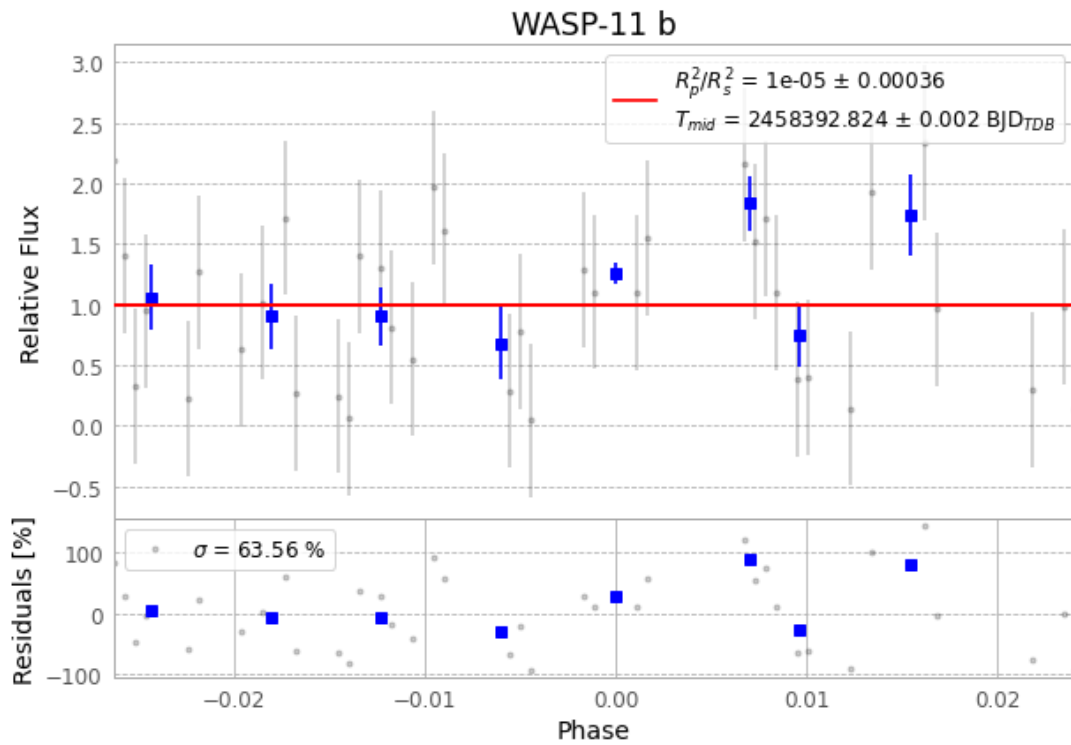


Figure 1 — FinalLightCurve_WASP-11 b_2018-09-30.png (SHA-256 901bc58f47f8a6a7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [295, 84], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9e1719f9fe77cac5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

93 FITS frames (61 MB), HATP-10181001051814.FITS → HATP-10181001095412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-181001.log (a488e1b6e539...), FinalLightCurve_WASP-11 b_2018-09-30.png (901bc58f47f8...), FinalLightCurve_WASP-11 b_2018-09-30.csv (d9f7f54adf01...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 01:42Z.